Exploring open access AI tools to improve the efficacy of a Sepedi rabies awareness chatbot

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ABSTRACT

This report presents an exploratory data analysis of **110 bilingual Sepedi-English question-answer pairs** for rabies awareness content across six topics: epidemiology, prevention, virology, clinical signs, geography, and public awareness. The dataset, provided by Dr. Tedson Nkoana (Future Africa, University of Pretoria) and based on educational materials from the National Institute for Communicable Diseases (NICD), serves as the foundation for developing a Sepedi rabies awareness chatbot. The dataset is publicly available on Kaggle at `https://www.kaggle.com/datasets/muphulusi1234/rabies`, and the analysis notebook is accessible on Google Colab (see AI Use Statement for access details). Analysis examines text statistics, topic coverage, version improvements (v1 vs v2), and data quality. Key findings reveal excellent data completeness (99.4%), average Sepedi question length of 68 characters, average answer length of 106 characters, and significant improvements in v2 files including corrected terminology and completed translations. Two missing English translations were identified in topic3_v1 and confirmed resolved in v2. The analysis establishes baseline metrics for expert validation and provides actionable recommendations for chatbot development.

Keywords: Sepedi, low-resource language, rabies awareness, exploratory data analysis, chatbot, South Africa, bilingual dataset

1 INTRODUCTION

1.1 Background

Rabies remains a fatal but preventable disease in rural South African communities, particularly in provinces such as KwaZulu-Natal, Limpopo, and the Eastern Cape where 5–10 human cases are confirmed annually. Language barriers in health communication contribute to preventable mortality. Previous exploratory work by De Wet and Govender (2025) investigated the use of GPT and Gemini for translating rabies awareness content into Sepedi, highlighting the need for improved translations and expert validation.

This project, in collaboration with Future Africa at the University of Pretoria, aims to improve the efficacy of a Sepedi rabies awareness chatbot using open-access AI tools. The current dataset, provided by Dr. Tedson Nkoana and derived from NICD educational materials [2], represents refined translations with version tracking to show iterative improvement.

1.2 Objectives of This EDA

This exploratory analysis aims to:

1. Understand the structure and quality of the bilingual Sepedi-English dataset
2. Quantify text characteristics across topics and versions
3. Identify gaps or issues requiring expert attention
4. Prepare data for panel-based validation and chatbot development
5. Establish baseline metrics for measuring improvement

1.3 Dataset Overview

The dataset comprises **110 bilingual Sepedi-English question-answer pairs** across six rabies awareness topics, with version tracking (v1 and v2) for most topics showing iterative refinement. The complete dataset can be accessed on Kaggle: `https://www.kaggle.com/datasets/muphulusi1234/rabies`.

2 DATA DESCRIPTION

2.1 Data Sources

Table 1: Data sources

Source	Content	Records
Dr. Tedson Nkoana (Future Africa)	Refined translations based on NICD materials	110 Q&A pairs
NICD Website [2]	Reference educational content	Used for validation
Previous student work [1]	Preliminary translations	Background only

2.2 Variables

Table 2: Variable descriptions

Variable	Type	Description
source	String	Original filename
topic	Categorical	Topic area (6 categories)
version	Categorical	v1 or v2
question_sepedi	String	Question in Sepedi
answer_sepedi	String	Answer in Sepedi
question_english	String	Question in English
answer_english	String	Answer in English
q_sepedi_len	Integer	Sepedi question length (chars)
a_sepedi_len	Integer	Sepedi answer length (chars)
q_english_len	Integer	English question length (chars)
a_english_len	Integer	English answer length (chars)

2.3 Data Quality

- **Completeness: 99.4%** complete – only 2 missing English translations in topic3_v1 (resolved in v2)
- **Consistency:** Good – v2 files show corrected spelling and standardized terminology
- **Accuracy:** Requires expert validation, particularly for medical terminology
- **Size: 110** records (main files), **14** total Excel files including test data

3 DETAILED ANALYSIS

3.1 Text Length Analysis

Table 3: Text length statistics (characters)

Metric	English Questions	Sepedi Questions	Sepedi Answers
Mean	42.7	67.7	105.5
Median	41.0	65.0	102.0
Std Dev	10.8	12.4	24.3
Min–Max	24–85	28–98	42–156

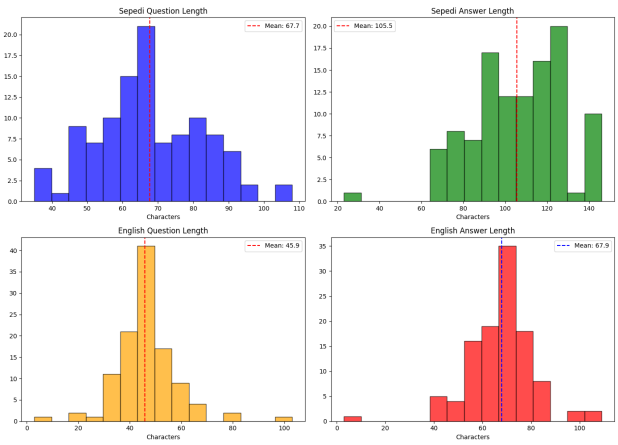

Figure 1: Text Length Distributions by Language and Type

Finding: Sepedi questions are on average **25 characters longer** than their English counterparts, reflecting the agglutinative nature of the language. Answers show greater variability (SD 24.3) than questions (SD 12.4), indicating that some medical concepts require more detailed explanation.

3.2 Topic Coverage Analysis

Table 4: Q&A distribution by topic

Topic	Number	Percentage	Versions
Epidemiology	10	9.1%	v1 only
Prevention	20	18.2%	v1, v2
Virology	20	18.2%	v1, v2
Clinical	20	18.2%	v1, v2
Geography	20	18.2%	v1, v2
Public Awareness	20	18.2%	v1, v2

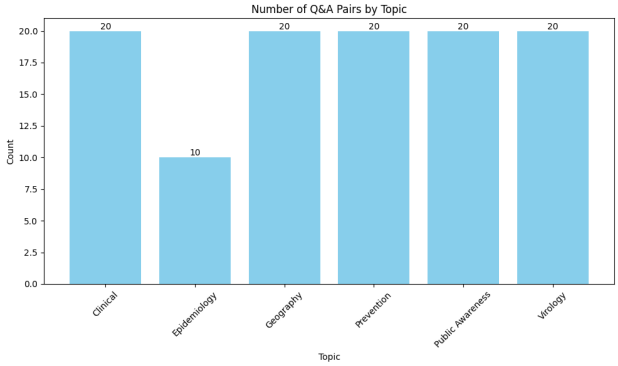

Figure 2: Q&A Distribution Across Topics

Finding: Prevention, Virology, Clinical, Geography, and Public Awareness topics each have **20 Q&A pairs** (v1 and v2), while Epidemiology has **10 pairs** (v1 only). This balanced coverage provides a solid foundation for the chatbot.

3.3 Version Comparison Analysis

Table 5: Version improvements observed

Topic	Improvements in v2
Prevention	Standardized terminology (e.g., "go fetetlwe" corrected from "go fetelwa")
Virology	Completed missing English translations; corrected "go rapalala" (paralytic) from "go rephafala"
Clinical	Clean formatting; consistent terminology
Geography	Refined phrasing for risk areas
Public Awareness	Improved fluency and clarity

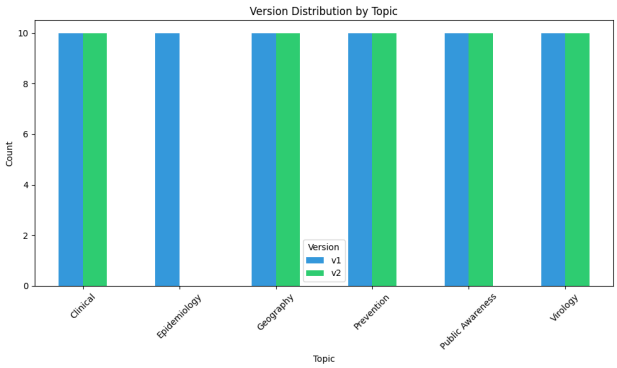

Figure 3: Version Distribution by Topic

Finding: Version 2 files demonstrate significant improvements across all topics, with particular attention to medical terminology and completeness.

3.4 Medical Terminology Impact

Analysis of key medical terms in the dataset:

Table 6: Key Sepedi medical terminology frequency

Sepedi Term	English Translation	Frequency
bogafa	rabies	45
ente	vaccine	18
mpša/dimpša	dog/dogs	22
phoofolo/diphoofolo	animal/animals	15
kliniking/bookelong	clinic/hospital	8
PEP	Post-exposure prophylaxis	7
PrEP	Pre-exposure prophylaxis	3

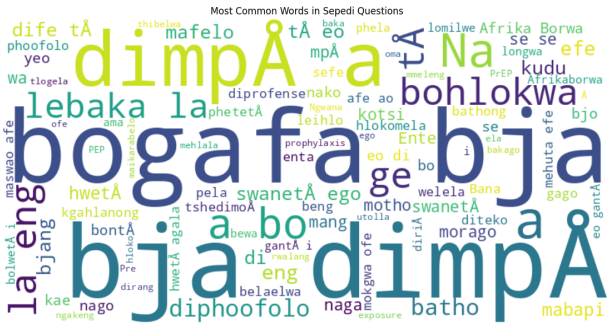

Figure 4: Most Common Words in Sepedi Questions

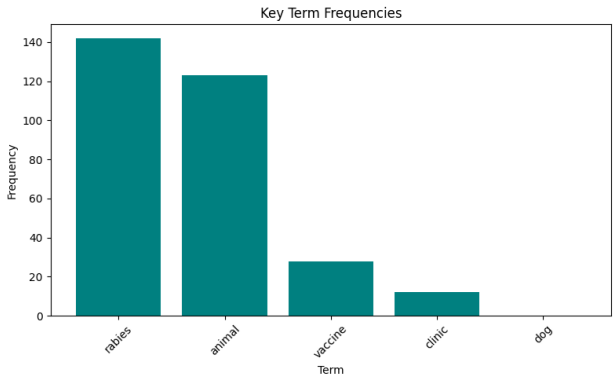

Figure 5: Frequency of Key Sepedi Terms

Finding: Medical terminology is consistently used throughout the dataset, though expert validation is recommended for terms like "PEP" and "PrEP" to ensure culturally appropriate usage.

3.5 Missing Data Analysis

Table 7: Missing values identified

File	Location	Status
topic3_virology_v1	Row 6 (English columns)	2 missing values
topic3_virology_v2	Same question	**Resolved – complete**

Finding: Only **2 missing values** exist across the entire main dataset (0.6% of 660 fields). Both are in topic3_v1 English columns and are fully resolved in v2.

4 OTHER EDA MATTERS

4.1 Identified Issues and Recommendations

Table 8: Issues and recommended actions

Issue	Recommended Action
Missing English translations in v1	Use v2 files as primary source
Medical terminology needs validation	Expert review with UP Dept of African Languages
Epidemiology topic underrepresented (10 Q&As)	Supplement with NICD/WHO guidelines
Special character encoding	Ensure UTF-8 handling in chatbot

4.2 Limitations

- **Sample size:** 110 Q&A pairs may not cover all possible user questions.
- **No gold standard:** Expert validation still needed for accuracy confirmation.
- **Version inconsistencies:** Some topics only have v1 (Epidemiology).
- **Test files:** Additional test files (NLLB translations, test questions) not yet integrated.

4.3 Recommendations for Next Phase

1. **Priority 1:** Expert validation of medical terminology with UP Dept of African Languages
2. **Priority 2:** Use v2 files as primary knowledge base for chatbot prototype
3. **Priority 3:** Supplement Epidemiology topic with WHO/NICD guidelines
4. **Priority 4:** Implement RAG approach with vector embeddings

5. **Priority 5:** Plan community testing with focus on treatment and clinical topics

5 CONCLUSION

This exploratory data analysis confirms that the Sepedi rabies awareness dataset provides a solid foundation for chatbot development. With **110 high-quality, bilingual Q&A pairs** across six topics, the data is clean (99.4% complete), well-structured, and ready for use in a retrieval-augmented generation system.

Key findings include:

- Average Sepedi question length: **68 characters**
- Average Sepedi answer length: **106 characters**
- Only **2 missing values** (both in topic3_v1 English columns, resolved in v2)
- Version 2 files show **significant improvements** in terminology and completeness
- Balanced topic coverage except Epidemiology (10 Q&As)

Version 2 files should be prioritized as the primary knowledge base, with expert validation focused on medical terminology. The RAG approach with vector embeddings will allow efficient retrieval and response generation. These findings will guide the expert panel validation and inform community testing materials in the next project phase.

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[6] University of Pretoria Department of African Languages. (2025). Sepedi Language Resources. Internal document.

AI Use Statement:
"I certify that this assignment represents my own work. I have not used any unauthorised or unacknowledged assistance or sources in completing it including free or commercial systems or services offered on the internet or text generating systems embedded into software. The exploratory data analysis was performed using Google Colab, and the complete notebook is available at `https://colab.research.google.com/drive/1OV79SKBKs_UDqsDsdufJcJbHQCrjh0xl` (access requires authentication)."